

38,117

ACTIVE SOURCE LOC

Dalam direktori src/

33,541

PURE CODE LINES

Eksklusif komentar & spasi

149

CORE SOURCE FILES

100% TypeScript (0 JSX)

45

FUNCTIONAL MODULES

21 App Router Endpoints

Ringkasan Eksekutif Arsitektur

Sistem **Dashboard BVL v2** beroperasi dengan stack enterprise modern berbasis **Next.js 16.2.4 (Turbopack)**, **React 19.2.4**, **Supabase PostgreSQL Cloud**, dan **Tailwind CSS v4**. Seluruh basis kode telah termigrasi 100% ke TypeScript murni dengan 71 static/dynamic routes tervalidasi lolos build produksi tanpa error.

1. DISTRIBUSI BAHASA & EKSTENSI BERKAS (CORE WORKSPACE)

FORMAT / BAHASA	KATEGORI FUNGSIONAL	FILES	TOTAL BARIS	PURE CODE	KOMENTAR	PROPORSI
.tsx TypeScript React	Komponen Antarmuka (UI) & Page Routes	74	26,303	23,753	328	54.5%
.ts TypeScript Logic	Business Services, Context, API Endpoints	80	11,793	9,769	513	22.4%
.json JSON Config	Package manifests & compiler settings	5	8,539	8,536	0	19.6%
.md Dokumentasi	SOP Operasional, Panduan Integrasi, Log	12	1,196	753	185	1.7%
.sql SQL Migrations	Supabase DDL, Trigger Audit & RLS Policies	8	399	307	48	0.7%
.js Maintenance Scripts	Utility & operational runners	2	380	307	10	0.7%
.css CSS Stylesheet	Tailwind v4 theme variables & globals	1	170	126	15	0.3%
TOTAL WORKSPACE (EXCL. LOCKFILES & BACKUP DUMPS)		182	48,780	43,551	1,099	100.0%

2. DISTRIBUSI ARSITEKTUR LAYER (DIREKTORI SRC/)

LAYER ARSITEKTUR	TANGGUNG JAWAB TEKNIS	FILES	TOTAL BARIS	PURE CODE	% CODE
src/app/	Next.js App Router (45 Route Modul + 21 API Endpoints)	102	29,589	26,404	78.7%
src/services/	Business Logic Layer, Supabase Aggregations, Data Services	22	5,728	4,727	14.1%
src/components/	Reusable Global UI Widgets (KPI Cards, Charts, Modals)	13	2,097	1,899	5.7%
src/utils/	Supabase Client/Server Configuration & Middleware Guard	4	313	229	0.7%
src/lib/	Global Context (RBAC), Formatters & Security Tools	6	298	218	0.6%
src/scripts/ & proxy	Migration runners & edge proxy	2	92	64	0.2%
TOTAL AKTIF DI src/		149	38,117	33,541	100.0%

3. BREAKDOWN MODUL BISNIS UTAMA (TOP FEATURE MODULES)

NAMA MODUL	CAKUPAN FUNGSIONALITAS & DELIVERABLES	FILES	TOTAL BARIS	TINGKAT MODULARITAS
Invoice Management	Header-Detail accordion, detail modal, 5-sheet Executive Excel Report	12	3,174	Tinggi (Sangat Bersih)
API Endpoints (/api)	Backend services, auth verify, sync cron, customer data access	21	2,027	Modular (REST)
Mobile Interface (/m)	Dedicated responsive view: CRM profiling, daily sales, advisor list	7	1,746	Modular (Sub-routes)
Monthly Transactions	Virtual pagination data grid, multi-store filtering, item details	6	1,529	Modular Component
CRM Deduplication	Algoritma matching no. HP, email normalizer, merging tool	3	1,430	Terkonsolidasi
Crossing Sales	Cross-store crossing matrix, advisor allocation, Excel export builder	7	1,286	Modular (Reports Split)
Advisor Performance	Target vs achievement, KPI ranking, individual advisor tracking	1	1,166	Monolitik Page
Daily Sales Report	Laporan harian store, multi-year comparison, instant PDF export	1	1,054	Monolitik Page

4. ANALISIS BERKAS TERBESAR & POTENSI REFAKTORISASI

BERKAS SUMBER	BARIS	TIPE & KOMPLEKSITAS	REKOMENDASI PEMECAHAN / TINDAKAN
src/app/advisor-performance/page.tsx	1,166	Tinggi (> 1,000 LOC)	Pecah ke subkomponen: AdvisorTable, KPICards, dan FilterBar.
src/app/daily-report/page.tsx	1,054	Tinggi (> 1,000 LOC)	Ekstraksi export logic (jsPDF/XLSX) ke utility terpisah.
src/app/daily-breakdown/page.tsx	1,039	Tinggi (> 1,000 LOC)	Pisahkan visual chart dan data summary table.
src/services/reportService.ts	939	Sedang (Multi-domain)	Split per domain (misal: crossingReportService & monthlyReportService).
src/app/sales-simulator/page.tsx	936	Sedang (Simulation UI)	Pindahkan simulation math formulas ke custom hook useSalesSimulator().

5. KESIMPULAN & REKOMENDASI ROADMAP

KEKUATAN ARSITEKTUR SAAT INI

- Zero Build Errors:** 71 halaman tervalidasi build Next.js 16 tanpa peringatan tipe.
- Pola Modular Baru Berhasil:** Modul invoice-management menjadi standar emas pemisahan UI, modal, dan sheet builder.
- Audit Trail Aktif:** Trigger PostgreSQL mencatat riwayat perubahan data krusial.
- Dual Persistence:** Sales journal memiliki sinkronisasi Supabase + fallback offline.

REKOMENDASI TAHAP BERIKUTNYA

- Modularisasi Halaman Besar:** Pecah 3 halaman > 1,000 baris menggunakan pola sub-komponen.
- Unified Export Utility:** Satukan fungsi PDF/Excel export agar tidak menduplikasi styling antar modul.
- Optimasi Payload Query:** Implementasi server-side caching untuk endpoint transaksi volume tinggi.